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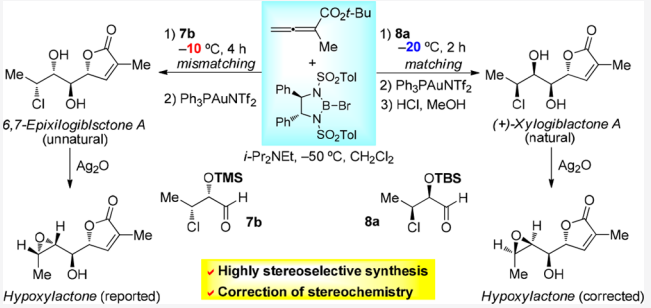
Synthesis of (+)-Hypoxylactone through Allenoate γ-Addition: Revision of Stereochemistry

Gyungah Pak, Euijin Park, Saehansaem Park, and Jimin Kim*

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ABSTRACT: A synthesis of (+)-hypoxylactone has been accomplished in four steps starting from the allenoate γ-addition of *threo*-3-chloro-2-silyoxybutanals, leading to the revision of stereochemistry. The key was the discovery of control elements required to matching/mismatching cases in the allenoate γ-addition to provide the desired adducts as a single isomer. The utility of the γ-adduct was demonstrated with the Au(I)-catalyzed cyclization to afford (+)-xylogiblactone A. Use of Ag_2O was the key to epoxidation for preventing epimerization of the γ-lactone ring.



Target-oriented synthesis (TOS) becomes an important concept to provide natural products and their derivatives in an efficient way.[1] Due to scarcity in nature, TOS is crucial to increase access to natural products often identified as bioactive substances. This protocol can supply both natural and unnatural compounds in quantity and allow continuing biological studies to unveil new properties and functions.[2] Ideally, concise syntheses with perfect regio- and stereoselectivity providing a single product offer efficient and practical tools to achieve the ultimate goal.

Structurally unique γ-lactone polyketide (+)-hypoxylactone (**1**) was isolated from the fermentation broth of marine fungus *Hypoxylon croceum* found in a mangrove estuary drift wood along with several sordarins including hypoxysordarin (**2**).[3] The new sordarin derivatives enable selective inhibition of protein synthesis, resulting in antifungal activity. In the plausible biological pathway, the initial step with the side chain in hypoxysordarin gives an acetal enzyme complex through a nucleophilic addition of a hydrolase onto the ester. The release of hypoxylactone together with sordarin can be catalyzed by the hydrolase through an epoxide opening event as illustrated in Figure 1. The structure of hypoxylactone (**1**) was deduced by comparison with NMR data of co-occurring hypoxysordarin (**2**). To the best of our knowledge, only similar polyketides from natural butenolides[4] are antifungal xylogiblactones A, B, and C isolated from a different fermented broth of *Xylotunulus gibbisporus* YMJ863 in 2014.[5] In this respect, it is possible that the side chain bearing a *cis*-epoxide in hypoxylactone (**1**) could be derived from a chlorohydrin unit in xylogiblactone A (**4**) or vice versa (Figure 1).

In view of stereochemical connection for both natural γ-lactone polyketides, we were intrigued by the fact that hypoxylactone (**1**) correlates with non-natural xylogiblactone

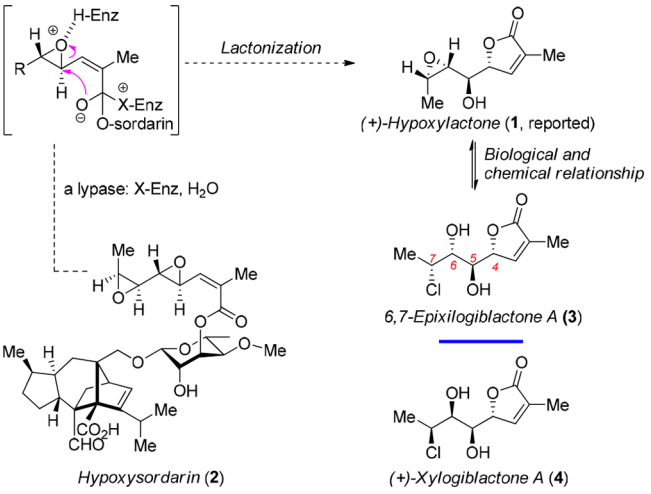
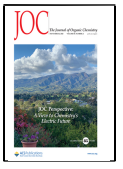


Figure 1. (+)-Hypoxylactone and related substances.

A (**4**), but its epimer, unnatural 6,7-*epi*-xylogiblactone A (**3**) in the biological or chemical phenomenon. This could be defined as an unusual case in spite of the different origins of natural sources for **1** and **4**.

Allene has been proved to be a useful functionality for a variety of transition-metal catalysis, particularly for cyclizations in the construction of carbo- and heterocycles.[6] Recently, we disclosed

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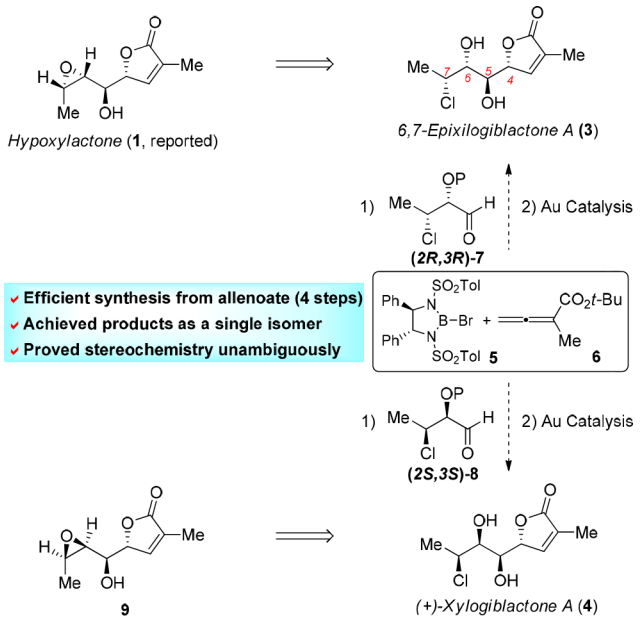
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a novel procedure of asymmetric allenoate γ-addition reactions to provide γ-adducts in the construction of axial and center chirality simultaneously.[7] The γ-adduct is readily amenable for further conversion to heterocyclic synthetic intermediates including γ-lactones by gold catalysis.[8] Concise synthesis of (+)-xylogiblactone A (**4**) through the allenoate γ-addition of unprecedented kinetic resolution of racemic aldehydes clearly demonstrates the efficacy and applicability of this methodology.[9] The characteristic features of this approach, in terms of the chemical efficiency of the unusual stereoselective assembly and the structural features of the products, have prompted us to carry out further investigations as depicted in Scheme 1. It was

Scheme 1. Synthetic Strategy

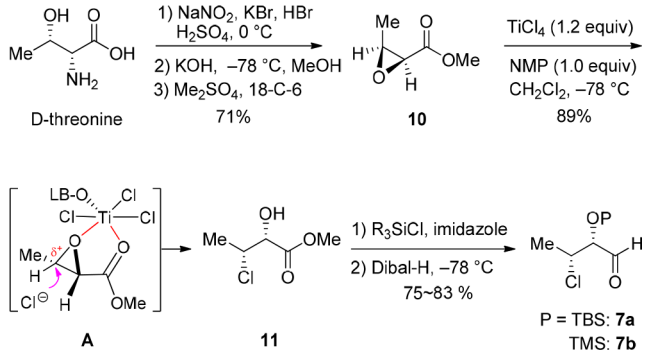


envisaged that syntheses of **3** and **4**, potential precursors for hypoxylactone, could be realized in a predictable fashion by the allenoate γ-additions with chiral aldehydes **7** and **8**.

Herein, we wish to describe concise syntheses of **3** and **4**, demonstrating the utility of the allenoate γ-addition reaction and the correlation between two natural products, hypoxylactone (**1**) and xylogiblactone A (**4**). This investigation led to the correction and verification of the known relative and absolute stereochemistry of hypoxylactone (**1**).

With this issue in mind, we began with readily available L- and D-threonines for the preparation of starting chiral aldehydes **7** and **8** in quantity, as modified and improved literature procedure (Scheme 2).[10] First, the *cis*-epoxide **10** was obtained in three steps from D-threonine: the transformations first involved diazotization and bromination of D-threonine. The resulting bromohydrine was converted to an epoxidation under basic conditions, and then the carboxylate was treated with Me_2SO_4 to produce the desired epoxy ester **10**. Early trials for the epoxide opening reaction of **10** with $TiCl_4$ or $SnCl_4$ under various conditions[11] indicated that the conversion to the desired chlorohydrin **11** could not be satisfied presumably due to a lack of reactivity. Although the reaction produced **11**, it suffered from low yield (13%) and several impurities.[12] We subsequently considered that a Lewis base catalyst, such as HMPA, DMF, or DMP, previously utilized for a variety of transformations through increasing reactivity of Lewis acid,[13] could be employed

Scheme 2. Preparation of Chiral Aldehydes 7a and 7b



for this transformation. Fortunately, utilization of NMP (*N*-methylpyrrolidone, 1.0 equiv) with $TiCl_4$ (1.2 equiv) at −78 °C cleanly provided **11** in 89% yield as a single isomer, not only increasing the reactivity but also improving chemical yields via **A** in Scheme 2. The resulting ester **11** was converted to the aldehyde **7a** through a reduction using Dibal-H (2.1 equiv) at −78 °C for 30 min after the protection of a hydroxyl group by TBS.

The stage was thus set for the γ-addition reaction of **6** with the aldehydes **7** or **8** mediated by chiral bromoborane **5**.[14] Initial attempts of the allenoate γ-addition of **6** to **7a** (P = TBS) could not be realized under various conditions. The reaction of **6** with (*R*,*R*)-**5**, prepared *in situ* from (1*R*,2*R*)-1,2-diphenylethane-1,2-bis-tosylsulfonamide and BBr_3, in the presence of *i*-Pr_2NEt (2.1 equiv) at −50 °C for 30 min in CH_2Cl_2, followed by the addition of **7a** at −20 °C indicated that the conversion to the γ-adduct completely failed, owing to a lack of reactivity caused by a geometrical issue. This is not a surprising outcome based on the analysis of the reaction pathway depicted in Scheme 3.

In fact, the low reactivity can be predictable by stereoelectronic effects and geometrical preference: Stereochemical model **B**, considered as the Felkin−Anh model, could be electronically favored due to an interaction of the $\pi^*_{C=O}$ orbital with an adjacent perpendicular σ^*_{C-O} orbital, lowering the LUMO energy level.[15] However, a severe steric interaction in model **B** exists between R and Me, leading to a mismatching pair in the stereochemical interaction.[16] We subsequently speculated that the modification of **7a** might be required to reduce steric bulkiness. Accordingly, the TBS group was replaced with a smaller TMS group. After surveying several conditions with **7b**, several key findings emerged: (1) an enolization time for 30 min at −50 °C was crucial for optimal conditions (Table 1, entry 4 and entries 1~4); (2) reactions performed at −10 °C in CH_2Cl_2 resulted in the best chemical yields in comparison with other solvents, including toluene and a mixed solvent system (Table 1, entries 7 and 8). Although **12** was produced during the reaction, the low chemical yield (11%) remained to be improved (Table 1, entry 3). Fortunately, we were delighted to find an optimal condition, which is the reaction of **6** with **5a** in the presence of *i*-Pr_2NEt at −50 °C for 30 min, and then the addition of **7b** at −50 °C warming up to −10 °C for 4 h to give the γ-adduct **12** in 34% isolated yield as a single regio- and stereoisomer (Table 1, entry 4).

Although the exact mechanistic aspects of this transformation have not been rigorously elucidated, Scheme 3 illustrates possible stereochemical routes **B** and **C** for the γ-allenoate addition adduct. Due to a severe steric repulsion between R and Me in electronically favored the Felkin−Anh model **B**, model **B**

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Scheme 3. Completion of a Synthesis of Reported Hypoxylactone (1)

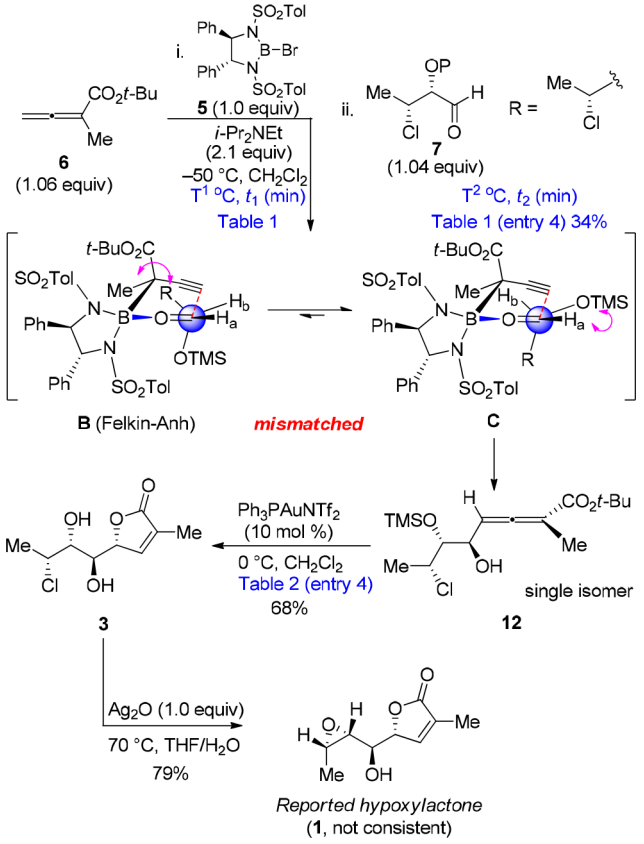


Table 1. Orienting Investigations for γ-Addition of 6 with 7b to 12[a]

entry	T^1 °C/t_1 (min)	T^2 °C/t_2 (h)	solvent	yield (%)[c]	ee (%)[d]
1	−78/60	−10/3	CH_2Cl_2	NR	
2	−50/60	−20/4	CH_2Cl_2	<10	
3	−50/30	−20/4	CH_2Cl_2	11	>99
4	−50/30	−10/4	CH_2Cl_2	34	>99
5	−50/20	−10/3	CH_2Cl_2	21	>99
6	−30/30	−0/3	CH_2Cl_2	18	>99
7	−50/30	−10/4	toluene	<5	NA
8	−50/30	−10/3	CH_2Cl_2/$PhCF_3$[b]	<10	>99

[a]Reactions were performed under a dried N_2 atmosphere. [b]A mixture of 4:1. [c]Isolated yields (%) after purification by flash chromatography. [d]Enantiomeric excess (%) values were determined by NMR analysis.

could be converted to the more sterically stable model **C**. We believe that the origin of the stereochemical outcomes for **12** via γ-addition might be a subtle geometrical preference for the orientation sterically less demanding in model **C**.

It is worth noting that γ-adduct **12** is a key intermediate to construct the reported structure of hypoxylactone (**1**). Next, a regiospecific oxacyclization of **12** was conducted under the conditions of gold catalysts to form the γ-lactone moiety (Table 2). After investigations for optimizing the cyclization, the use of $Ph_3PAuNTf_2$ proved to be superior to other gold catalysts (Table 2, entries 4 and 5). The cyclization of **12** in the presence of $Ph_3PAuNTf_2$ (10 mol %) at 0 °C for 1 h in CH_2Cl_2 took place to provide the desired TMS deprotected 6,7-*epi*-xylogiblacotone

Table 2. Oxacyclization of 12 to 3 via Gold Catalysis[a]

entry	catalyst	mol %	solvent	T °C/t (h)	yield (%)[d]
1	Ph_3AuCl	10	CH_2Cl_2	30/7	<5
2	Ph_3AuBF_4[b]	10	CH_2Cl_2	20/5	41
3	Ph_3AuAsF_6[b]	10	CH_2Cl_2	20/5	34
4	$Ph_3PAuNTf_2$[c]	10	CH_2Cl_2	0/1	68
5	$Ph_3PAuNTf_2$	10	$PhCF_3$	20/5	55
6	$Ph_3PAuNTf_2$	5	CH_2Cl_2	20/7	47

[a]Unless otherwise stated, all reactions were performed on a 0.1 M solution under a dried N_2 atmosphere. [b]Prepared *in situ* by stirring of a 1:1 mixture of Ph_3PAuCl and AgX (X = BF_4 or AsF_6). [c]Tf = trifluorosulfonyl ($-SO_2CF_3$). [d]Isolated yield (%) after purification by flash chromatography.

A (**3**, Table 2, entry 4). The structure of **3** was unambiguously confirmed by X-ray crystallography as shown in Figure 2.

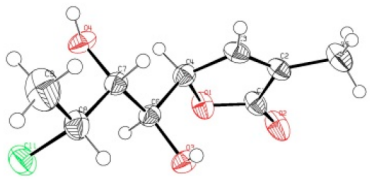


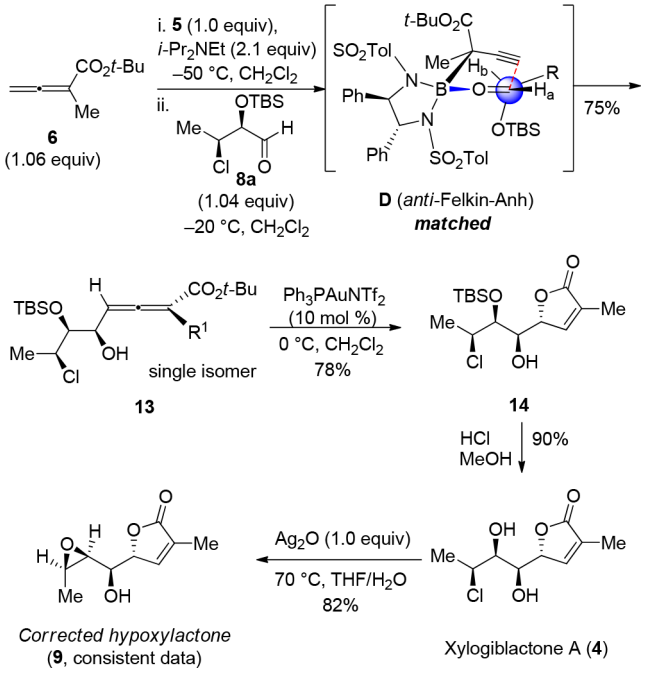
Figure 2. X-ray structure of **3** (ellipsoid contour 50% probability level).

The final epoxidation of a chlorohydrin unit in **3** was conducted under various basic conditions. However, reactions always produced **1** along with a mixture of inseparable diastereomers mainly due to an epimerization at C4 of the γ-lactone unit.[17] Epoxidations at low temperatures turned out to be inefficient in terms of chemical conversion and purity in the desired product. After intensive investigations for selective epoxidation, it was gratifying to find that the use of Ag_2O (1.0 equiv) in aqueous THF at 70 °C for 4 h proved to be optimal to form the reported hypoxylactone (**1**) without producing the epimer.[18] Finally, NMR data of **1** was compared with the reported data, but observed data did not match the reported values.[3] The comparison results could suggest structural incorrection of hypoxylactone (**1**) in the previous report.

In light of the above results that the stereoselective synthesis of **1** turned out not to be a natural diastereomer of hypoxylactone, we next drew attention to the application of our approach for the synthesis of the natural hypoxylactone on the basis of a hypothetical biological relationship with xylogiblactone A (**4**). We undertook the synthesis of **9** starting from aldehyde **8a** (R = TBS), prepared from the previous sequence for **7**, as shown in Scheme 4. The allenoate γ-addition reaction of **8a** under similar conditions afforded the γ-adduct **13** as a single regio- and stereoisomer in 75% yield. The reaction underwent smoothly even at a lower temperature, −20 °C, after an enolization of the allenoate **6** at −50 °C for 30 min. This result shows a clear case of the matched reaction pair. That is, reactivity and selectivity of the reaction can arise from the matched case, anti-Felkin−Anh model **D**, favored in both electronic and steric factors. With a possible connection between (+)-xylogiblactone A (**4**) and hypoxylactone, γ-adduct **13** was carried out for a Au(I)-catalyzed (10 mol %) cyclization to form the lactone **14** in 78% yield. Next, the TBS deprotection of **14** was initially attempted with a basic fluoride source, but the γ-lactone moiety turned out to be sensitive to the basic fluoride causing epimerization at the C4 position of γ-lactone as

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Scheme 4. Synthesis and Correction of Natural (+)-Hypoxylactone (9)



previously observed. After examining various conditions, the TBS group was cleanly taken off with 10% aq HCl (5.5 equiv) in MeOH at rt to result in (+)-xylogiblactone A (**4**).

Then, the resulting (+)-xylogiblactone A (**4**) was treated with Ag_2O (1.0 equiv) in aqueous THF at 70 °C to provide the epoxide **9** as a sole product. Finally, we were delighted to find that all physical data of **9** including spectroscopic data were identical with the original data.[3] This outcome supports that the stereochemical assignment of the epoxide in hypoxylactone needs to be revised from **1** to **9**. The result clearly shows chemical and biological connections between hypoxylactone (**9**) and xylogiblactone A (**4**).

In summary, we demonstrated not only facile syntheses of (+)-xylogiblactone A (**4**) and hypoxylactone (**9**) by means of the allenoate γ-additions but also structural revision of hypoxylactone (**1**). According to the newly confirmed structure of the natural γ-lactone **9**, it also suggests that further verification of the structure of hypoxysordarin (**2**) will be necessary. Stereochemistry of (+)-hypoxylactone is revised unambiguously by the synthesis of both (+)-xylogiblactone A (**4**) and its epimer **3**. Further studies are in progress for synthetic applications to useful substances and evaluation of biological activities on natural and unnatural γ-lactones.

EXPERIMENTAL SECTION

General. Unless otherwise stated, reactions were run in flame-dried glassware under an atmosphere of nitrogen or argon. Solvents were dried by passage through an activated alumina column under argon. Also, dichloromethane was distilled from P_2O_5 prior to use. All liquid reagents purchased from Aldrich or Alfa Aesar were distilled properly prior to use, unless otherwise indicated. Triethylamine and diisopropylethylamine were distilled from CaH_2 prior to use. Purification was conducted by flash column chromatography on silica gel (230–400 mesh), eluting with a mixture of hexanes and ethyl acetate, unless otherwise stated. The reported yields refer to chromatographically purified and isolated products. All reactions were monitored by thin-layer chromatography carried out on Merck silica gel plate (60 F_{254}) using UV light as a visualizing agent and ethanolic anisaldehyde solution and heat as the developing agent. Silica gel 60 (TA792685, 230–400 mesh) from Merck was used for column chromatography. The reported yields are for chromatographically pure isolated products. ^1H and ^{13}C NMR spectra recorded on a Varian Unity Inova (at 300 and 75 MHz, respectively), Bruker Advance-500 (at 500 and 125 MHz, respectively), and/or Agilent ProPulse 500 (at 500 and 125 MHz, respectively) are reported relative to residual proton solvent ($CDCl_3$ = 7.26 and 77.0 ppm, respectively). Data for ^1H NMR spectra are reported as follows: chemical shift (δ ppm) (multiplicity, coupling constant in Hz, integration). NMR data were acquired at the Korea Basic Science Institute (KBSI, Gwangju branch). IR spectra were recorded on a Bruker Vertex 70 spectrometer and are reported in the frequency of absorption (cm^{-1}). High-resolution mass spectra (HRMS) were acquired from the Korea Basic Science Institute (KBSI, Daegu and Ochang branches) by using EI, FAB, or ESI mode and from Chonnam Center for Research Facilities (CCRF) with a JEOL JMS-T200GC by using field desorption method (FD). X-ray crystallography was performed by Dr. H. J. Lee at the Korea Basic Science Institute (KBSI, Western Seoul branch).

All reported values are within 0.5% of the calculated value. Optical rotations were measured on a PerkinElmer 343 plus at an ambient temperature (Na line).

Preparation of (2*S*,3*S*)-Methyl-3-methyloxirane-2-carboxylate (10). Methyl epoxybutanoate **10** was prepared from the modified literature procedure.[17] To a mixture of L-threonine (2 g, 16.8 mmol) and KBr (7.03 g, 59.1 mmol) in aq H_2SO_4 (1.25 M, 35 mL, 43.7 mmol) was added $NaNO_2$ (1.88 g, 27.2 mmol) in portions over 30 min at 0 °C. After stirring at the same temperature for 2 h, the resulting mixture was extracted with EtOAc three times. The combined organic layers were washed with distilled water, followed by brine, and the organic layer was dried over Na_2SO_4. After evaporation, the crude bromoacid was carried out for the epoxidation without further purification. The resulting crude mixture was dissolved in absolute MeOH (28 mL) and cooled down at −30 °C. Then, a solution of KOH (1.98 g, 33.6 mmol) in absolute MeOH (9.3 mL) was slowly added into the cold solution of the crude bromoacid. After stirring for 1 h, the reaction was warmed to rt and stirred for 12 h. The resulting suspension was filtered off, and then, the filtrate was concentrated down under a vacuum. The crude salt was taken onto the methylation without further purification. To a mixture of the crude salt and 18-C-6 (3.95 g, 14.9 mmol) in CH_2Cl_2 (21 mL) was slowly added Me_2SO_4 (1.4 mL, 14.9 mmol) at rt. After 1 h, Et_2O was added into the reaction mixture to precipitate out the $EtSO_4K$–crown ether complex. The resulting suspension was filtered off, and the volatile filtrate was carefully concentrated down under a vacuum. The crude mixture was purified by column chromatography (pentane/Et_2O = 8:1) to give the *cis*-epoxy ester **10** (1.38 g, 11.9 mmol, 71% in 3 steps) as a colorless oil. The NMR data were identical to those of the reported **10**:[19] R_f = 0.48 (3:1 hexane/EtOAc); $[\alpha]_D^{20}$ +12.1 (*c* 0.5, $CHCl_3$); IR (neat) 3500, 2922, 1742, 1441, 1380, 1276, 1139, 1041, 978 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 3.79 (s, 3H), 3.51 (d, 1H, J = 4.5 Hz), 3.29 (qd, 1H, J = 5.4, 4.5 Hz), 1.38 (d, 3H, J = 5.4 Hz); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 168.7, 53.5, 52.9, 52.3, 12.9; HRMS (EI+) m/z $[M]^+$ calcd for $C_5H_8O_3$ 116.0473, found 116.0476.

Preparation of (2*R*,3*R*)-Methyl-3-chloro-2-hydroxybutanoate (11). To a mixture of **10** (2.2 g, 18.9 mmol) and NMP (1.81 mL, 18.9 mmol) in CH_2Cl_2 (1.3 M, 14 mL) was added dropwise $TiCl_4$ (1.0 M in CH_2Cl_2, 23.4 mL, 23.44 mmol) at −78 °C. After 1.5 h, the reaction was quenched with 7% aq $NaHCO_3$ (50 mL) and stirred for 30 min at rt. Then, the resulting mixture was extracted with CH_2Cl_2 twice, and the combined organic layers were washed with brine. After drying with $MgSO_4$, the organic layer was concentrated down under a vacuum. The crude mixture was purified by column chromatography (hexane/EtOAc = 15:1 to 7:1) to afford **11**[20] as a colorless oil (2.58 g, 16.89 mmol, 89%): R_f = 0.45 (3:1 hexane/EtOAc); $[\alpha]_D^{20}$ −1.33 (*c* 0.5, $CHCl_3$); IR (neat) 2985, 1751, 1440, 1351, 1291, 1205, 1134, 1034, 855 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 4.44 (qd, 1H, J = 6.8, 1.9 Hz), 4.24 (dd, 1H, J = 7.8, 1.9 Hz), 3.82 (s, 3H), 3.12 (d, 1H, J = 7.8 Hz), 1.60 (d, 3H, J = 6.8 Hz); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 172.0, 74.1, 58.7, 53.0, 21.3; HRMS (EI+) m/z $[M]^+$ calcd for $C_5H_9ClO_3$ 152.0240, found 152.0239.

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Preparation of (−)-(2*R*,3*R*)-Methyl-3-chloro-2-((trimethylsily)oxy)butanoate (11a). To a mixture of **11** (2.58 g, 16.89 mmol) and imidazole (1.27 g, 18.58 mmol) in CH_2Cl_2 (1.0 M, 16.9 mL) was added dropwise TMSCl (2.37 mL, 18.58 mmol) at 0 °C. The reaction mixture was slowly warmed to rt and stirred for 30 min. The precipitate was filtered off and washed with CH_2Cl_2. The filtrate was evaporated under a vacuum, and the crude mixture was purified by column chromatography on silica gel (hexane/EtOAc = 55:1 to 30:1) to afford **11a** (3.14 g, 13.97 mmol, 83%) as a colorless oil: R_f = 0.60 (10:1 hexane/EtOAc); $[\alpha]_D^{20}$ −31.58 (*c* 0.5, $CHCl_3$); IR (neat) 2956, 1761, 1438, 1289, 1251, 1165, 1054, 993, 840 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 4.37 (qd, 1H, *J* = 6.8, 3.8 Hz), 4.24 (d, 1H, *J* = 3.8 Hz), 3.78 (s, 3H), 1.51 (d, 3H, *J* = 6.8 Hz), 0.17 (s, 9H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 171.1, 75.9, 58.3, 52.3, 21.0, −0.2; HRMS (EI+) *m*/*z* $[M - CH_3]^+$ calcd for $C_8H_{17}ClO_3Si$ 209.0401, found 209.0400.

Preparation of (−)-(2*R*,3*R*)-3-Chloro-2-((trimethylsilyl)oxy)-butanal (7b). To a solution of **11a** (1.08 g, 4.79 mmol) in CH_2Cl_2 (0.16 M, 29 mL) was added dropwise a solution of Dibal-H (1.1 M in cyclohexane, 11 mL, 10 mmol) at −78 °C. The reaction mixture was stirred for 30 min and then quenched with MeOH at the same temperature. The resulting mixture was warmed to rt and treated with sat. potassium sodium tartrate. After filtration, the filtrate was extracted with CH_2Cl_2 twice, and the combined organic layers were washed with brine. The organic layer was dried over Na_2SO_4 and concentrated down under a vacuum. The resulting crude mixture was purified by column chromatography (hexane/EtOAc = 40:1 to 25:1) to afford **7b** (770 mg, 3.95 mmol, 82%) as a colorless oil: R_f = 0.41 (10:1 hexane/EtOAc); $[\alpha]_D^{20}$ −47.2 (*c* 0.5, $CHCl_3$); IR (neat) 3477, 2958, 2869, 1738, 1445,1379, 1253, 1086, 875, 753 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 9.63 (d, 1H, *J* = 1.0 Hz), 4.29 (qd, 1H, *J* = 6.8, 3.6 Hz), 4.03 (dd, 3H, *J* = 3.6, 1.0 Hz), 1.53(d, 3H, *J* = 6.8 Hz), 0.19 (s, 9H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 201.7, 79.9, 56.7, 20.8, −0.0; HRMS (EI+) *m*/*z* $[M]^+$ calcd for $C_7H_{15}ClO_2Si$ 194.0530, found 194.0532.

(−)-(3*S*,5*R*,6*R*,7*R*)-*tert*-Butyl-7-chloro-5-hydroxy-2-methyl-6-((trimethylsilyl)oxy)octa-2,3-dienoate (12). A flame-dried 20 mL Schlenk flask containing (+)-(1*R*,2*R*)-1,2-diphenyl-1,2-bis-*p*-toluenesulfonylamide (0.50 g, 0.96 mmol) was charged with freshly dried CH_2Cl_2 (7 mL) under a nitrogen atmosphere. The resulting mixture was cooled to 0 °C and treated with BBr_3 (freshly prepared 1 M solution in CH_2Cl_2, 1.0 mL, 1.0 mmol). The solution was stirred at 0 °C for 30 min, warmed to 25 °C followed by stirring for an additional 1.5 h, and then concentrated under a vacuum (1 mmHg) through a side neck of the Schlenk flask connected with a three-way regulating valve. Dryness of the vacuum line was maintained with a drying tube containing anhydrous $CaSO_4$ and two traps (NaOH pellets and cold trap at −78 °C). Freshly distilled CH_2Cl_2 (5 mL) was added and evaporated under a vacuum as mentioned above. Freshly distilled CH_2Cl_2 (3 mL) was added, and the homogeneous solution of (*R*,*R*)-**5** was cooled to −50 °C and treated dropwise with a mixture of *tert*-butyl-2-methylbuta-2,3-dienoate **6** (0.16 g, 1.02 mmol) and *i*-Pr_2NEt (0.35 mL, 0.26 g, 2.01 mmol) in CH_2Cl_2 (1 mL). The reaction mixture was allowed to proceed at −50 °C. After stirring for 30 min at −50 °C, a precooled solution of **7b** (194 mg, 1.0 mmol) in CH_2Cl_2 (1 mL) was added via a cannular needle along the wall of the flask, while keeping the same temperature. The reaction was allowed to warm to −10 °C and proceed for 4 h. The resulting reaction mixture was quenched by the addition of an aqueous buffer solution (pH 7, 7 mL) followed by CH_2Cl_2 (ca. 10 mL) to dissolve the white precipitate (bis-sulfonamide). The aqueous layer was extracted with CH_2Cl_2 (ca. 10 mL × 2). The combined organic extracts were washed with brine (1 ×), dried over anhydrous Na_2SO_4, filtered, evaporated, and taken up in ether (ca. 20 mL). The solution was cooled to 0 °C for 20 min to complete precipitation of (+)-(1*R*,2*R*)-1,2-diphenyl-1,2-bis-*p*-toluenesulfonylamide by filtration through a sintered glass funnel. The filtrate was concentrated under reduced pressure, and the crude mixture was purified by column chromatography (hexane/EtOAc = 30:1) to afford **12** (115 mg, 0.33 mmol, 34%) as a single isomer: R_f = 0.26 (7:1 hexane/EtOAc); $[\alpha]_D^{20}$ −1.43 (*c* 0.5, $CHCl_3$); IR (neat) 3420, 2967, 2854, 1961, 1704, 1456, 1369, 1289, 1251, 1123, 1004, 738 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 5.62 (dq, 1H, *J* = 7.6, 3.0 Hz), 4.31 (ddd, 1H, *J* = 7.6, 6.1, 4.2 Hz), 4.14 (qd, 1H, *J* = 6.8, 6.5 Hz), 3.74 (dd, 1H, *J* = 6.5, 4.2 Hz), 2.11 (d, 1H, *J* = 6.1 Hz), 1.86 (d, 3H, *J* = 3.0 Hz), 1.48 (d, 3H, *J* = 6.8 Hz), 1.46 (s, 9H), 0.20 (s, 9H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 208.8, 166.3, 99.2, 93.3, 81.3, 80.3, 70.4, 59.2, 28.1, 21.6, 15.1, 0.7; HRMS (FD+) *m*/*z* $[M + H]^+$ calcd for $C_{16}H_{30}ClO_4Si$ 349.1596, found 349.1597.

(−)-(*R*)-5-((1*S*,2*R*,3*R*)-3-Chloro-1,2-dihydroxybutyl)-3-methylfuran-2(5*H*)-one (3). To a flame-dried 10 mL round flask containing $Ph_3PAuNTf_2$ (30 mg, 0.019 mmol) was added a solution of **12** (67.5 mg, 0.19 mmol) in CH_2Cl_2 (2 mL) at 0 °C. The reaction was proceeded for 1 h at the same temperature, and then the resulting mixture was concentrated under reduced pressure. The crude mixture was purified by column chromatography (hexane/EtOAc = 3:1 to 1:1) to afford the lactone **3** (28 mg, 0.13 mmol, 68%) as a white solid: R_f = 0.31 (1:1 hexane/EtOAc); $[\alpha]_D^{20}$ −53.3 (*c* 0.5, $CHCl_3$); IR (neat) 3473, 2985, 2903, 1737, 1447, 1373, 1236, 1096, 938, 787 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 7.15−7.17 (m, 1H), 5.26 (ddq, 1H, *J* = 4.0, 1.9, 1.9 Hz), 4.59 (qd, 1H, *J* = 6.8, 2.6 Hz), 4.05 (dd, 1H, *J* = 8.5, 4.0 Hz), 3.55 (dd, 1H, *J* = 8.5, 2.6 Hz), 1.96 (dd, 3H, *J* = 1.9, 1.7 Hz), 1.66−1.74 (brs, 1H), 1.62 (d, 3H, *J* = 6.8 Hz), 1.61−1.62 (brs, 1H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 174.0, 145.1, 131.9, 81.2, 75.2, 71.4, 60.0, 21.7, 10.9; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_9H_{13}ClO_4Na$, 243.0400, found 243.0402.

(+)-Hypoxylactone (1, Reported). To a solution of **3** (20 mg, 0.091 mmol) in THF (2.3 mL) and H_2O (0.6 mL) was added Ag_2O (21 mg, 0.091 mmol) at rt. The mixture was stirred at 70 °C in an oil bath for 4 h and then cooled to rt. The resulting suspension was filtered through a short Celite pad, and the filtrate was concentrated down under a vacuum. The crude mixture was purified by column chromatography on SiO_2 (hexane/EtOAc = 3:1) to provide compound **1**[21] (13 mg, 0.072 mmol, 79%): R_f = 0.3 (1:1 hexane/EtOAc); $[\alpha]_D^{20}$ +23.7 (*c* 0.2, $CHCl_3$); IR (neat) 3430, 2928, 1747, 1208, 1050, 790 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 7.17 (s, 1H), 5.07−5.10 (m, 1H), 3.89 (dd, 1H, *J* = 7.9, 3.6 Hz), 3.15−3.20 (m, 1H), 2.91 (dd, 1H, *J* = 7.9, 4.0 Hz), 1.98 (s, 3H), 1.96 (brs, 1H), 1.40 (d, 3H, *J* = 5.5 Hz); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 173.8, 144.5, 131.9, 82.2, 68.5, 54.7, 52.6, 13.3, 10.9; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_9H_{12}O_4Na$ 207.0633, found 207.0633.

Preparation of (2*S*,3*S*)-Methyl-2-((*tert*-butyldimethylsilyl)-oxy)3-chlorobutanoate (*ent*-11a). Compound *ent*-**11** was prepared from L-threonine according to the procedure for **11**. To a solution of *ent*-**11** (600 mg, 3.95 mmol) in DMF (0.4 M, 10 mL) were added imidazole (538 mg, 7.9 mmol) and TBSCl (1.11 g, 7.11 mmol), and the reaction mixture was stirred for 15 h at rt. The reaction mixture was diluted with brine and extracted with EtOAc. The organic layer was dried over $MgSO_4$ and evaporated to give a crude oil, which was purified by column chromatography on SiO_2 (hexane/EtOAc = 40:1 to 20:1) to afford the ester *ent*-**11a** (1.05 g, 3.93 mmol, 99%) as a colorless oil. The NMR data were identical with those of the reported racemic ester:[20] R_f = 0.52 (20:1 hexane/EtOAc); $[\alpha]_D^{20}$ +32.1 (*c* 0.5, $CHCl_3$); IR (neat) 2900, 2810, 1765, 1447, 1373, 1096, 980 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 4.28−4.36 (m, 1H), 4.25 (d, 1H, *J* = 4.0 Hz), 3.76 (s, 3H), 1.52 (d, 3H, *J* = 6.7 Hz), 0.93 (s, 9H), 0.11 (s, 3H), 0.08 (s, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 171.2, 76.3, 58.5, 52.1, 25.7, 21.1, 18.4, −4.9, −5.3; LC−MS *m*/*z* $[M + Na]^+$ calcd for $C_{11}H_{23}ClO_3SiNa$, 289.1, found 289.0.

Preparation of (2*S*,3*S*)-2-((*tert*-Butyldimethylsilyl)oxy)-3-chlorobutanal (8a). To a solution of *ent*-**11a** (1.05 g, 3.93 mmol) in CH_2Cl_2 (0.16 M, 25 mL) was added dropwise a solution of Dibal-H (1.1 M in cyclohexane, 7.4 mL, 8.17 mmol) at −78 °C. The reaction mixture was stirred for 30 min and then quenched with MeOH at the same temperature. The resulting mixture was warmed to rt and treated with sat. potassium sodium tartrate. After filtration, the filtrate was extracted with CH_2Cl_2 twice, and the combined organic layers were washed with brine. The organic layer was dried over Na_2SO_4 and concentrated down under a vacuum. The resulting crude mixture was purified by column chromatography (hexane/EtOAc = 40:1 to 25:1) to afford **8a** (781 mg, 3.30 mmol, 84%) as a colorless oil. The NMR data were in agreement with the reported data:[20] R_f = 0.41 (20:1 hexane/EtOAc); $[\alpha]_D^{20}$ +3.2 (*c* 0.5, $CHCl_3$); IR (neat) 2910, 2810, 1737, 1470, 1156, 1096, 938, 825 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 9.64 (d, 1H,

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J = 1.3 Hz), 4.26 (qd, 1H, J = 6.8, 3.7 Hz), 4.03 (dd, 1H, J = 3.7, 1.3 Hz), 1.54 (d, 3H, J = 6.8 Hz), 0.95 (s, 9H), 0.11 (s, 3H), 0.10 (s, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 201.8, 80.0, 56.9, 25.7, 20.9, 18.3, −4.6, −5.1; GC−MS m/z $[M - COH]^+$ calcd for $C_9H_{20}ClOSi$ 207.1, found 207.1.

(+)-(3*S*,5*R*,6*S*,7*S*)-*tert*-Butyl-6-((*tert*-butyldimethylsilyl)oxy)-7-chloro-5-hydroxy-2-methylocta-2,3-dienoate (13). A flame-dried 20 mL Schlenk flask containing (+)-(1*R*,2*R*)-1,2-diphenyl-1,2-bis-*p*-toluenesulfonylamide (0.50 g, 0.96 mmol) was charged with freshly dried CH_2Cl_2 (7 mL) under a nitrogen atmosphere. The resulting mixture was cooled to 0 °C and treated with BBr_3 (freshly prepared 1 M solution in CH_2Cl_2, 1.0 mL, 1.0 mmol). The solution was stirred at 0 °C for 30 min, warmed to 25 °C followed by stirring for an additional 1.5 h, and then concentrated under a vacuum (1 mmHg) through a side neck of the Schlenk flask connected with a three-way regulating valve. Dryness of the vacuum line was maintained with a drying tube containing anhydrous $CaSO_4$ and two traps (NaOH pellets and cold trap at −78 °C). Freshly distilled CH_2Cl_2 (5 mL) was added and evaporated under a vacuum as mentioned above. Freshly distilled CH_2Cl_2 (3 mL) was added, and the homogeneous solution of (*R*,*R*)-**5** was cooled to −50 °C and treated dropwise with a mixture of *tert*-butyl-2-methylbuta-2,3-dienoate **6** (0.16 g, 1.02 mmol) and *i*-Pr_2NEt (0.35 mL, 0.26 g, 2.01 mmol) in CH_2Cl_2 (1 mL). The reaction mixture was allowed to proceed at −50 °C. After stirring for 30 min at −50 °C, a precooled solution of **8a** (237 mg, 1.0 mmol) in CH_2Cl_2 (1 mL) was added via a cannular needle along the wall of the flask while keeping the same temperature. The reaction was allowed to warm to −20 °C and proceed for 2 h. The resulting reaction mixture was quenched by the addition of an aqueous buffer solution (pH 7, 7 mL) followed by CH_2Cl_2 (ca. 10 mL) to dissolve the white precipitate (bis-sulfonamide). The aqueous layer was extracted with CH_2Cl_2 (ca. 10 mL × 2). The combined organic extracts were washed with brine (1×), dried over anhydrous Na_2SO_4, filtered, evaporated, and taken up in ether (ca. 20 mL). The solution was cooled to 0 °C for 20 min to complete precipitation of (+)-(1*R*,2*R*)-1,2-diphenyl-1,2-bis-*p*-toluenesulfonylamide by filtration through a sintered glass funnel. The filtrate was concentrated under reduced pressure, and the crude mixture was purified by column chromatography (hexane/EtOAc = 30:1) to afford **13** (281 mg, 0.72 mmol, 75%) as a single isomer: R_f = 0.30 (15:1 hexane/EtOAc); $[\alpha]_D^{20}$ +41.1 (c 0.3, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 5.42 (dd, 1H, J = 7.6, 2.9 Hz), 4.47 (dd, 1H, J = 7.6, 4.1 Hz), 4.14 (qd, 1H, J = 6.9, 3.7 Hz), 3.70 (dd, 1H, J = 4.1, 3.7 Hz), 2.34 (brs, 1H), 1.86 (d, 3H, J = 2.9 Hz), 1.53 (d, 3H, J = 6.9 Hz), 1.47 (s, 9H), 0.95 (s, 9H), 0.16 (s, 3H), 0.15 (s, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 209.2, 166.4, 99.0, 95.1, 81.0, 78.4, 69.8, 58.3, 28.1, 25.9, 20.9, 18.3, 14.9, −4.1, −4.3; LC−MS m/z $[M + Na]^+$ calcd for $C_{19}H_{35}ClO_4SiNa$, 413.2, found 413.1.

(+)-(*R*)-5-((1*R*,2*S*,3*S*)-2-((*tert*-Butyldimethylsilyl)oxy)-3-chloro-1-hydroxybutyl)-3-methylfuran-2(5*H*)-one (14). To a flame-dried 10 mL round flask containing $Ph_3PAuNTf_2$ (30 mg, 0.019 mmol) was added a solution of **13** (74.3 mg, 0.19 mmol) in CH_2Cl_2 (2 mL) at 0 °C. The reaction was proceeded for 1 h at the same temperature, and then the resulting mixture was concentrated under reduced pressure. The crude mixture was purified by column chromatography (hexane/EtOAc = 3:1 to 1:1) to afford the lactone **14** (49.6 mg, 0.15 mmol, 78%) as a white solid. Physical data are in agreement with the reported data:[20] R_f = 0.27 (5:1 hexane/EtOAc); $[\alpha]_D^{20}$ +31.2 (c 0.7, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.33 (dq, 1H, J = 1.7, 1.7 Hz), 4.60 (ddq, 1H, J = 9.3, 1.7, 1.7 Hz), 4.16 (d, 1H, J = 4.8 Hz), 4.02 (qd, 1H, J = 6.9, 4.8 Hz), 3.64 (d, 1H, J = 9.2 Hz), 1.95 (dd, 3H, J = 1.7, 1.7 Hz), 1.71−1.75 (brs, 1H), 1.51 (d, 3H, J = 6.9 Hz), 0.94 (s, 9H), 0.20 (s, 3H), 0.18 (s, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 173.8, 148.3, 130.5, 79.8, 73.2, 70.8, 57.9, 25.8, 19.5, 18.1, 10.7, −4.2, −4.6; LC−MS m/z $[M + Na]^+$ calcd for $C_{15}H_{27}ClO_4SiNa$ 357.1, found 357.1.

(+)-Xylogiblactone A (4). Compound **14** (30.6 mg, 0.091 mmol) was dissolved in 1 mL of MeOH; 10% aq HCl (0.16 mL, 0.50 mmol) was added, and the mixture was stirred at rt overnight. The reaction was quenched with saturated aq $NaHCO_3$, and the resulting mixture was extracted with CH_2Cl_2. The combined organic layers were washed with brine and dried over Na_2SO_4. After concentration under reduced pressure, the crude compound was purified by flash column chromatography (hexane/EtOAc = 3:1) to afford (+)-xylogiblactone A (**4**) (18 mg, 0.082 mmol, 90%). Physical data were identical with those of the reported (+)-xylogiblactone A:[22] R_f = 0.25 (1:1 hexane/EtOAc); $[\alpha]_D^{20}$ +62.8 (c 0.6, MeOH); IR (neat) 3388, 2923, 2853, 1736, 1458, 1017, 879 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 7.33 (dq, 1H, J = 1.8, 1.8 Hz), 4.93 (ddq, 1H, J = 7.5, 1.8, 1.8 Hz), 4.24 (dq, 1H, J = 6.7, 6.7 Hz), 3.84 (dd, 1H, J = 7.5, 1.5 Hz), 3.61 (dd, 1H, J = 6.7, 1.5 Hz), 2.23−2.64 (brs, 2H), 1.95 (dd, 3H, J = 1.8, 1.8 Hz), 1.56 (d, 3H, J = 6.7 Hz); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 173.8, 147.6, 130.9, 80.5, 74.4, 72.1, 61.3, 21.0, 10.7; LC−MS m/z $[M + Na]^+$ calcd for $C_9H_{13}ClO_4Na$ 243.1, found 243.1.

(+)-Hypoxylactone (9, Corrected). To a solution of **4** (20 mg, 0.091 mmol) in THF (2.3 mL) and H_2O (0.6 mL) was added Ag_2O (21 mg, 0.091 mmol) at rt. The mixture was stirred at 70 °C in an oil bath for 4 h and then cooled to rt. The resulting suspension was filtered through a short Celite pad, and the filtrate was concentrated down under a vacuum. The crude mixture was purified by column chromatography on SiO_2 (hexane/EtOAc = 3:1) to provide compound **9** (13.7 mg, 0.075 mmol, 82%). Physical data were identical with those of the reported (+)-hypoxylactone:[21] R_f = 0.25 (1:1 hexane/EtOAc); $[\alpha]_D^{20}$ +153.7 (c 0.2, $CHCl_3$); IR (neat) 3436, 2920, 1751, 1208, 1060, 835 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) δ 7.28 (dq, 1H, J = 1.8, 1.8 Hz), 4.86 (ddq, 1H, J = 7.8, 1.8, 1.8 Hz), 3.44−3.52 (m, 1H), 3.26 (qd, 1H, J = 5.6, 4.3 Hz), 3.17 (dd, 1H, J = 6.1, 4.3 Hz), 2.36 (d, 1H, J = 4.4 Hz), 1.96 (dd, 3H, J = 1.8, 1.8 Hz), 1.36 (d, 3H, J = 5.6 Hz); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 173.5, 146.5, 131.2, 81.0, 70.1, 57.5, 53.9, 13.7, 10.8; HRMS (FD+) m/z $[M + H]^+$ calcd for $C_9H_{13}O_4$ 185.0808, found 185.0809.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c02194.

Full spectroscopic data of new and known compounds and crystal structures of **3** (PDF)

Accession Codes

CCDC 2017001 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: + 44 1223 336033.

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Author Contributions

The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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